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High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System

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***Abstract*— This paper presents a monostatic backscatter communication system that introduces a novel approach for backscatter multimedia data communication (BMC) in low-power Internet of Things environments. To enable an extremely low-cost and low-power implementation, an Arduino platform was employed as the switch for On-Off Keying (OOK) modulation. The system achieves a data rate of 80 kbps and maintains a Signal-to-Noise Ratio (SNR) of approximately 30 dB at a 30 cm distance between the tag and the reader, ensuring stable and reliable communication. Furthermore, a planar 2×1 Yagi-Uda antenna incorporating Split-Ring Resonators (SRRs) was designed to achieve high gain (7 dBi) and excellent antenna-to-antenna isolation (below -30 dB) within the ISM band (2.4~2.5 GHz). The significance of this work lies in developing an end-to-end backscatter communication system that successfully implements the transmission, reception, and reconstruction of complex image data using custom hardware and data processing algorithms.**

I. Introduction

The rapidly evolving Internet of Things (IoT) ecosystem demands energy-efficient and cost-effective communication systems to support a myriad of connected devices. Traditional wireless approaches often rely on active transmissions, which require significant power and complex hardware—factors that pose major limitations for battery operated or energy harvesting IoT devices. In contrast, backscatter communication systems present a highly attractive alternative, reflecting carrier waves and modulated signals rather than generating their own RF signal. This technique dramatically reduces energy consumption, enabling simpler, and more lightweight deployments that are particularly advantageous for large-scale IoT networks.

Recent research has expanded the scope of backscatter communication beyond basic sensor readings (e.g. temperature or humidity) to accommodate more demanding data types, such as multimedia. For instance, in smart factories, real-time video feeds monitor production lines and ensure worker safety, and in agriculture, unmanned camera systems capture crop and livestock images to optimize resource management. These advanced applications highlight the need for increased data throughput and reliability in low-power systems [1].

Within this backscatter paradigm, the system is designed as a monostatic system, where the transmitter and receiver are spatially co-located, to simplify the overall system requirements. By combining low-power backscatter communication with conventional modulation schemes, specifically On-Off Keying (OOK) modulation, robust data transmission can be achieved at a practical data rate of 80 kbps. In this work, an Arduino Uno microcontroller, widely known for its accessibility and educational use, was utilized as the tag, while a Software-Defined Radio (SDR) served as the reader. This configuration results in a highly cost-effective and straightforward solution. Performance in the ISM band (2.4~2.5 GHz) is further enhanced by integrating advanced antenna technologies, such as planar Yagi-Uda designs and SRRs. These antenna structures provide high gain and excellent isolation, which is essential for minimizing self-interference in monostatic systems and enabling reliable communication in energy-constrained IoT environments.

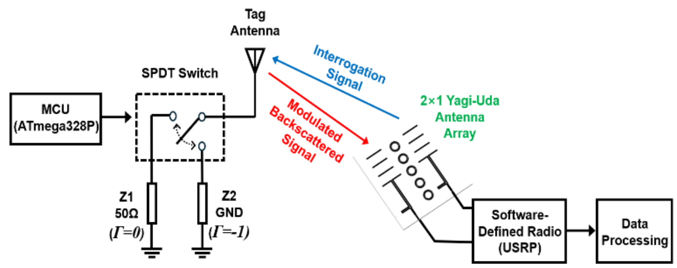


Figure 1. Block diagram of the proposed BMC system.

II. Measurement Setup

The backscatter communication system comprises two primary components: a tag and a reader. The tag modulates and scatters the carrier signal transmitted by the reader, and the

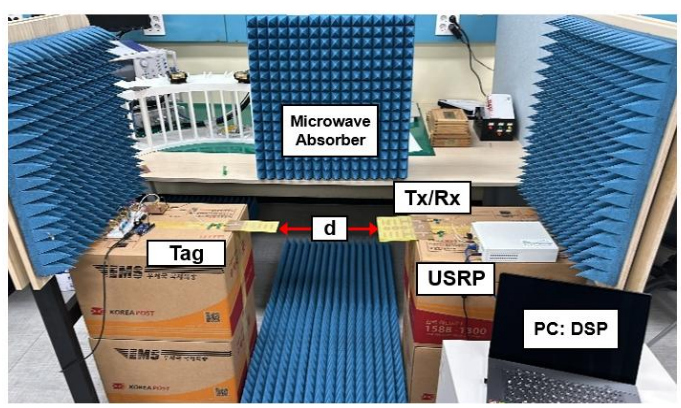
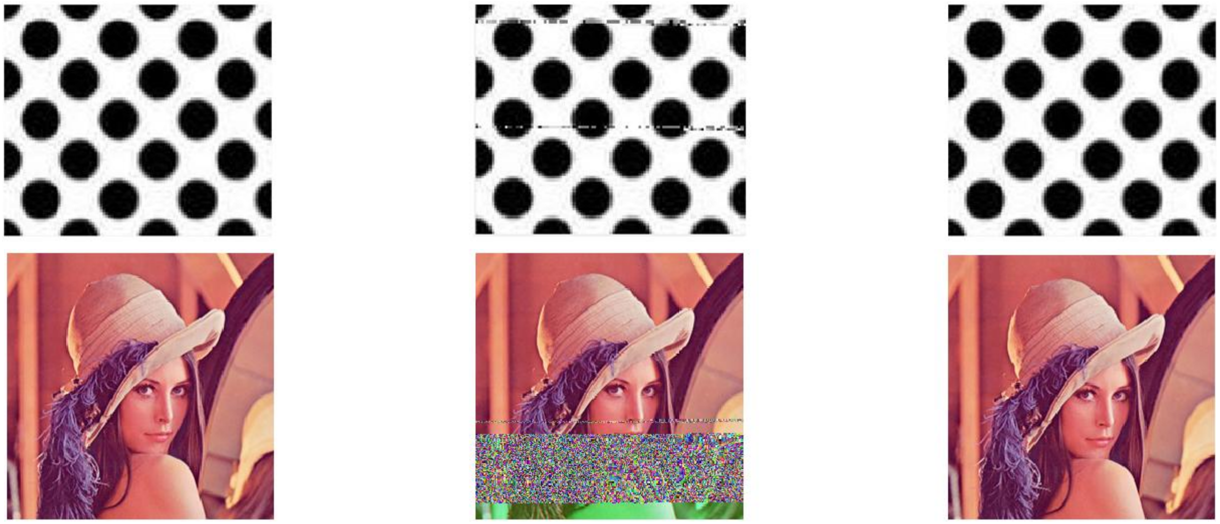


Figure 2. The Experimental setup of the proposed BMC system (d = 30 cm).



(a) (b) (c)

Figure 3. (a) Original, reconstructed images with (b) BER = 10^{-2} and (c) BER < 10^{-7} (no error bits).

scattered signal is subsequently received by the same reader. The block diagram of the proposed BMC system is shown in Figure 1. In this setup, the SDR, USRP N200, functions as the reader, generating a 2.45 GHz carrier wave and simultaneously serving as the receiver. The system utilized a monostatic architecture, wherein the RF source transmitted an incident signal to the tag, which modulated and reflected it back to the reader. The tag's microcontroller unit (MCU), an ATmega328P from the Arduino Uno, was controlled using dedicated software. The scattered signals received by the reader were analyzed through a data processing algorithm implemented in MATLAB.

To address the challenges associated with transmitting complex image data wirelessly—such as moderate transmission times and vulnerability to channel variations and multi-path interference—a microwave absorber (VHP NRL) was employed. It minimized dispersion and multi-path effects. For the experiments, grayscale dot images and the color version of the Lena image—commonly used in signal processing and educational contexts—were transmitted, with 8 bits per pixel for grayscale and 24 bits per pixel for color. The experimental setup of the proposed BMC system is shown in Figure 2.

III. Operation Principle And Analysis

Based on the control signal from the MCU, the SPDT switch generates two distinct signals by toggling between a 50 Ω load and ground. Each signal alters the load impedance of the antenna. The reflection coefficient, Γ, which is influenced by the antenna impedance (Z_a) and the load impedance (Z_L) of the tag, is given by the following expression:

$$\Gamma = \frac{Z_L - Z_a^*}{Z_L + Z_a} \tag{1}$$

Here, Z_a^* represents the complex conjugate of the antenna impedance. For OOK modulation, when Z_L is terminated with 50 Ω, the reflection coefficient becomes 0, corresponding to a non-reflection mode. Conversely, when Z_L is terminated with ground, the reflection coefficient becomes -1, representing a reflection mode. These modes are assigned to bits 0 and 1, respectively [2].

In designing the BMC system for the ISM band, the antenna plays a critical role in ensuring efficient transmission and reception. While the monostatic structure offers the advantage of space efficiency, it also introduces significant interference between the transmitted carrier wave and the received backscattered signals due to the overlap between transmission and reception channels. This interference greatly reduces the Signal-to-Noise Ratio (SNR), increasing the Bit Error Rate (BER) and making multimedia data communication more challenging.

To address these issues, two primary objectives should be met: (i) achieving high isolation to minimize mutual coupling between the electric fields in the transmission and reception paths, and (ii) attaining high gain to reliably detect weak backscattered signals. To meet these requirements, the system employs planar Yagi-Uda antennas with end-fire radiation patterns for both the tag and reader, ensuring optimal performance in the ISM band [3]. As shown in Figure 4, an array of SRRs is employed to suppress mutual coupling in the electric field. Figure 5 demonstrates that $|S_{21}|$ is measured to be less than -30 dB within the operating frequency range compared to -23 dB when SRRs are not integrated. Furthermore, as illustrated in Figure 6, the measured radiation patterns show a gain of 7 dBi, which closely matches the simulation result of 7.1 dBi. These experimental measurements demonstrate that the proposed custom-designed antenna configuration reduces

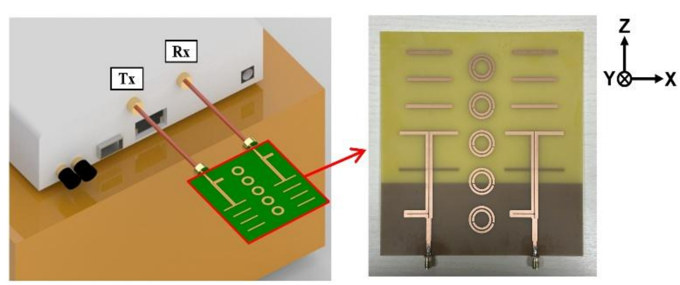


Figure 4. Geometry of the proposed Yagi-Uda antenna array with SRRs.

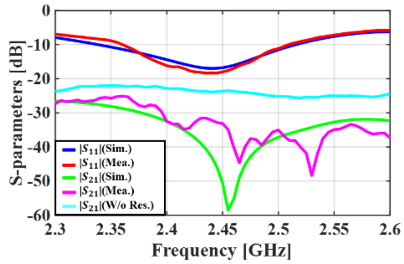


Figure 5. Measured antenna's S-parameters: $|S_{11}|$ & $|S_{21}|$

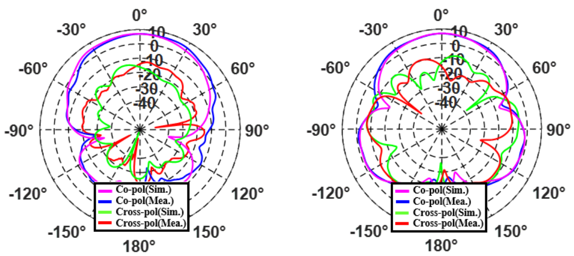


Figure 6. Measured gain radiation patterns at 2.45 GHz:
(a) $\emptyset = 0°$ (xz-plane), (b) $\emptyset = 90°$ (yz-plane).

electrical coupling, thereby improving the SNR and enhancing the precision of backscattered signal detection.

IV. Communication Demonstration

The Arduino Uno, with a limited SRAM capacity of only 2 kB, faces substantial constraints when attempting to process large or complex data (e.g. multimedia data) in a single operation. To address the hardware limitation during data transmission, large data sets are typically divided into smaller segments, such as 256 or 512 bytes, which are then processed sequentially. This approach ensures that each segment remains within the available memory capacity, thereby preventing overflow. After the data is received by the reader, it is processed using MATLAB software. The error probability for OOK modulation is expressed as follows [4]:

$$P_{e,o} = Q\left(\frac{d}{2\sigma}\right) = Q\left(\frac{\sqrt{2E_b}}{2\sqrt{N_o/2}}\right) = Q\left(\sqrt{\frac{E_b}{N_o}}\right) \tag{2}$$

Here, the Q-function is defined as $Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^{+\infty} e^{-t^2/2}\,dt$. In this equation, d is the distance between two constellation points, σ represents the standard deviation of the Gaussian noise. E_b is the average bit energy, and N_o is the noise power spectral density. For demodulation, the received real (I) and imaginary (Q) data were extracted and processed through amplitude peak detection based on their magnitudes. This approach was evaluated with two data sets: a black/white dot image (size: 70 k-bits) and a more complex color image (size: 1.6 M-bits), both transmitted at a data rate of 80 kbps.

Without the use of microwave absorbers, data transmission over a 30 cm distance yielded an SNR of approximately 13 dB and a BER of 10^{-2}, resulting in the reconstructed image shown in Figure 3(b). Given the 5 dB noise figure (*NF*) of the SDR system, further SNR enhancement was crucial to mitigate noise

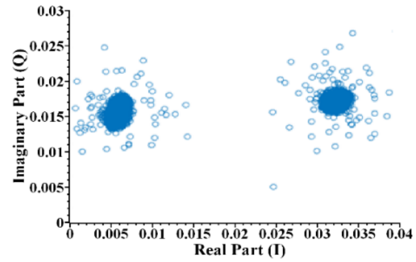


Figure 7. Constellation diagram of the received data.

and achieve error-free communication. Accordingly, microwave absorbers were deployed to minimize multi-path fading and ensure a focused Line-of-Sight (LoS) signal path. The absorbers effectively suppressed unwanted interference and reduced signal reflections, leading to a significant reduction in *No*, as indicated by (2). Consequently, the SNR improved from 13 dB to 30 dB at the same transmission distance of 30 cm. The enhanced SNR substantially reduced bit errors, as the higher SNR allowed the demodulator to accurately distinguish between binary states. Subsequent data processing algorithms confirmed the complete absence of errors, enabling reconstruction of the original image without error bits, as shown in Figure 3(c).

For accurate detection and optimal system performance, it is critical to eliminate DC offset caused by direct conversion in the hardware before demodulation. This offset can distort the received signal, displacing the constellation from its expected position and leading to demodulation inaccuracies [5]. The constellation diagram with the DC offset removed, processed using data processing algorithms, is shown in Figure 7.

V. Conclusion

The proposed BMC system employs monostatic architecture, and advanced antenna designs to enable efficient and reliable data transmission while minimizing power consumption and hardware complexity. By integrating an MCU with a digital switch as the tag and an SDR as the reader, the system achieves a cost-effective implementation. This design establishes a foundation for scalable IoT applications, with the potential to support a wide range of data types—from basic sensor readings to complex multimedia—thereby enhancing the functionality and efficiency of next-generation IoT networks, such as indoor communication applications.

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